



UBFC

UNIVERSITÉ
BOURGOGNE FRANCHE-COMTÉ



cea

Advancing the Verification and Error Analysis of Thermohydraulic Simulations in Nuclear Reactors

Master 2 CompuPhys - Internship Presentation | Ali FADLALLAH

Supervisors: Alexiane PLESSIER and Michael NDJINGA from DES/DM2S/STMF at CEA Saclay

28 August 2024



- 1 Introduction
 - Context
 - Objectives

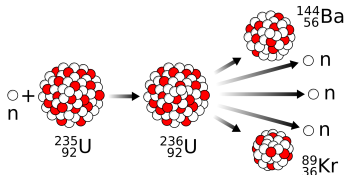
- 2 Methodology
 - Navier-Stokes equations
 - Error evaluation

- 3 Results
 - Single channel
 - Two parallel channels
 - Example via Jupyter Notebook

- 4 Conclusion and Perspectives

■ Nuclear energy

- **1938 Breakthrough:** Discovery of nuclear fission.
- **Chain Reaction:** Self-sustaining series of nuclear fission events releasing immense energy.
- **Historical Impact:** Led to the development of atomic bombs used in WWII.



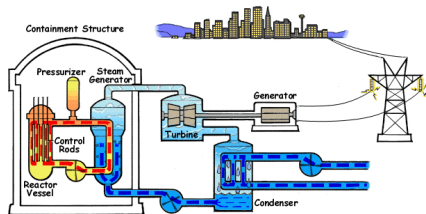
Chain reaction

■ CEA

- **Founded in 1945:** Established to harness and explore nuclear energy.
- **Leadership:** Pioneering advancements in nuclear system development.

■ Pressurized Water Reactor (PWR)

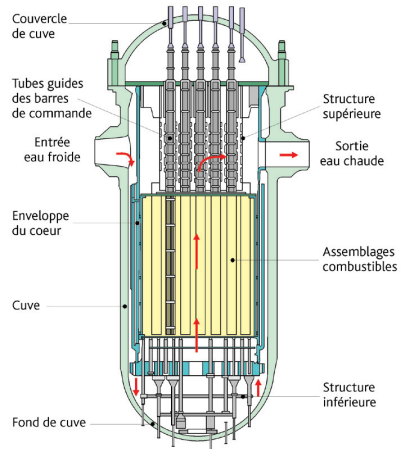
- **Global Prevalence:** The most widely used reactor type in nuclear power plants.
- **Cycle:** 3 circuits.
- **Core Function:** Nuclear fission generates heat, later converted into electrical power.
- **Coolant:** High-pressure water is used to transfer heat from the reactor core.



PWR



- Design phase and safety analysis
- Thermohydraulic numerical simulation software to model fluid dynamics.
- Focus on the fluid flow inside the reactor vessel
- Mathematical **verification** of the software for simple 1D geometries.



Reactor vessel



We have 3 main objectives for the internship:

- **Error Evaluation Scheme:** Develop a rigorous scheme for existing 1D single-channel flow cases.
- **New Test Case:** Introduce a new test case for two parallel channels.
- **Documentation Enhancement:** Use Jupyter Notebooks for improved clarity and usability.



- Fundamental equations governing fluid dynamics, they describes how a fluid evolves over time.
- For our purpose, we use the incompressible inviscid Navier-Stokes equations

$$\begin{cases} \nabla \cdot \mathbf{u} = 0, \\ \rho \frac{\partial \mathbf{u}}{\partial t} + \rho \nabla \cdot [\mathbf{u} \otimes \mathbf{u}] = -\nabla p + \mathbf{F}, \end{cases}$$

where, $\mathbf{u}$ is the fluid **velocity**, ρ is the density, p is the **pressure**, and $\mathbf{F}$ represents external forces.

- The software uses the **compressible** Navier-Stokes equations, the error evaluation is based on the **incompressible** Navier-Stokes equations.



- Variable of interest:

$$M = \frac{u}{c} \quad \text{is the Mach number}$$

For our cases we found that $M \approx 0.001$.

- The compressible regime is close to the incompressible one according to Schochet¹:

$$\begin{cases} u_{\text{comp}}(x) = u_{\text{incomp}}(x) + \mathcal{O}(M)(x), \\ p_{\text{comp}}(x) = p_{\text{incomp}}(x) + \mathcal{O}(M^2)(x). \end{cases}$$

- We proposed the following error evaluation scheme for low Mach flows:

$$\Delta f = \max \left\{ \left| \frac{f_{\text{software}}(x) - f_{\text{analytical}}(x)}{f_{\text{analytical}}(x) + \varepsilon} \right| \right\} < M$$

with f , velocity or pressure.

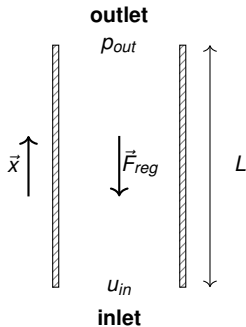
1. S. Schochet. Fast singular limits of hyperbolic pdes. Journal of Differential Equations, 114(2): 476–512, 1994. ISSN 0022-0396.



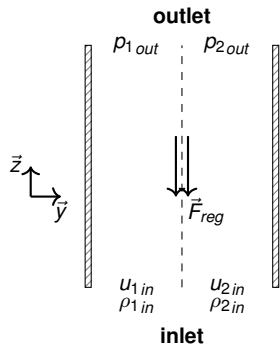
- Worked on 2 different type of cases for single and two channels involving different external forces and correlations such as:
 - Gravity
 - Pressure loss (regular friction)
 - Singular friction

- Simulate each physical phenomena individually

Two type of cases - regular friction



Single channel



Two parallel channels

$$\vec{F}_{reg} = -\frac{C_f}{2D_h} \rho ||\vec{u}|| \vec{u}$$

with:

C_f the friction coefficient (0.001 - 10)

D_h the hydraulic diameter (diameter of the channel)



- To run a case we write an input data file "jdd.data"
- We specify the mesh, the discretization scheme, and the solver.
- Set the boundary conditions and correlations such as:
 - Gravite
 - Diffusion
 - Temperature
 - u_{in} , p_{out}
 - Perte_Charge
- We run the case using Jupyter Notebook.
- We retrieve the numerical velocity and pressure profiles and compare them to the analytical ones.

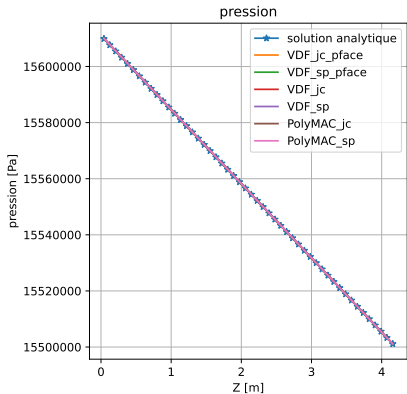
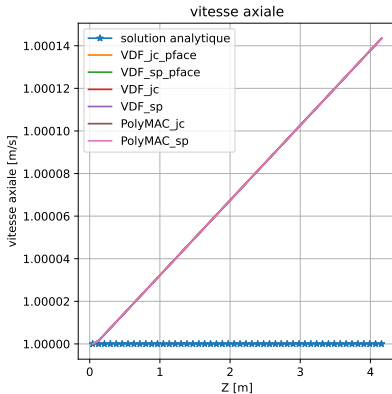
Input data file

```
78 lire pb
79 {
80     milieu_diphasique_1 {
81         gravite champ_uniforme 3 0. 0. 0.
82         loi_etat_diphasique Loi_StiffenedGas { }
83         Diametre_hyd_champ Champ_Uniforme 1 $Dh
84     }
85     Systeme_Diphasique_4eqs
86     {
87         convection_scal { amont }
88         convection_vit { amont_sans_porosite }
89         diffusion_scal { negligable }
90         diffusion_vit { negligable }
91         conditions_initiales {
92             pression                Champ_Uniforme 1 155.0e+05
93             concentration            Champ_Uniforme 1 0.0
94             enthalpie_liquide        Champ_Uniforme 1 1.3e6
95             enthalpie_vapeur         Champ_Uniforme 1 2.6e+06
96             vitesse_liquide          Champ_Uniforme 3 0. 0. 1.
97             vitesse_vapeur           Champ_Uniforme 3 0. 0. 0.
98         }
99         conditions_lignes
100         {
101             Zbas Entree_fluide_diphasique
102             {
103                 concentration        Champ_Front_Uniforme 1 0.
104                 enthalpie_liquide     Champ_Front_Uniforme 1 1.3e6
105                 enthalpie_vapeur      Champ_Front_Uniforme 1 2.6e+06
106                 vitesse_liquide       Champ_Front_Uniforme 3 0. 0. 1.
107                 vitesse_vapeur        Champ_Front_Uniforme 3 0. 0. 0.
108             }
109             Zhaut Sortie_libre_diphasique { pression Champ_Front_Uniforme 1 155e+05 }
110
111             Ybas Symetrie
112             Yhaut Symetrie
113             Xbas Symetrie
114             Xhaut Symetrie
115         }
116         sources
117         {
118             $frottement
119         }
120         modelisation
121         {
122             direction_principale_ecoulement 2
123             correlations
124             {
125                 Ecart_Vitesse_HOMOGENE
126             }
127         }
128     }
```

Single channel subject to pressure loss

- Analytical (incompressible) solution:

$$\begin{cases} u(x) = u_{in}, \\ p(x) = p_{out} - \frac{C_f}{2D_h} \rho u_{in}^2 (x - L). \end{cases}$$



Velocity and pressure profile for a single channel case with pressure loss



- **Recall:** that the criteria for success is the Mach number $M = 0.001$ or 0.1%

$$\Delta p = \max \left\{ \left| \frac{p_{\text{software}}(x) - p_{\text{analytical}}(x)}{p_{\text{analytical}}(x) + \varepsilon} \right| \right\} < 0.1\%$$

cases	Δp	$\Delta p_{\text{in-out}}$	Δu
PolyMAC_jc	0.00072 % ✓	0.0007 % ✓	0.014 % ✓
PolyMAC_sp	0.00072 % ✓	0.0007 % ✓	0.014 % ✓
VDF_jc	0.00072 % ✓	0.0007 % ✓	0.014 % ✓
VDF_jc_pface	0.00072 % ✓	0.0007 % ✓	0.014 % ✓
VDF_sp	0.00072 % ✓	0.0007 % ✓	0.014 % ✓
VDF_sp_pface	0.00072 % ✓	0.0007 % ✓	0.014 % ✓

Error table for the single channel with regular friction



- **Developed Error Evaluation Scheme:** Introduced a scientifically backed error evaluation scheme for 1D single-channel, single-phase flows.
- **New Two-Channel Flow Test Case:** Derived an analytical solution and proposed an evaluation scheme for the two parallel channels case.
- **Enhanced Documentation:** Improved the verification database by rewriting test cases in Jupyter Notebook.



- **Extend Verification to Two-Phase Flows:** Apply the new error evaluation methodology to two-phase flows, including defining relevant Mach numbers and ensuring convergence to the incompressible regime.
- **Enhance Two-Channel Testing:** Expand testing for two-channel scenarios by incorporating cases involving singular friction, viscosity, and heat conduction.
- **Update/Enrichment of the documentation of existing notebooks.**

UBFC

UNIVERSITÉ
BOURGOGNE FRANCHE-COMTÉ



Thank you for listening

CEA SACLAY

91191 Gif-sur-Yvette Cedex
France

alifadlallah.fr@gmail.com



- **Velocity:** $u = 1 \text{ m/s}$
- **Pressure:** $p = 1.55 \times 10^7 \text{ Pa}$
- **Temperature:** $T = 566 \text{ K}$

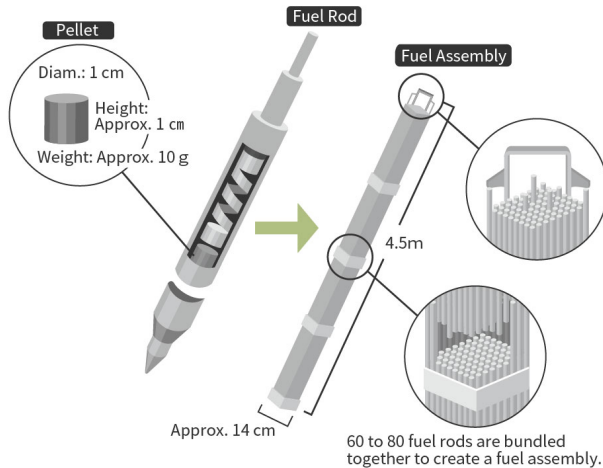
The International Association for the Properties of Water and Steam (IAPWS):

- **Density:** $\rho = 740 \text{ Kg/m}^3$
- **Dynamic viscosity:** $\mu \approx 9 \times 10^{-5} \text{ Pa} \cdot \text{s}$
- **Speed of sound:** $c_s = 1009 \text{ m/s}$

For a characteristic length $D_h = 1 \text{ m}$; we have:

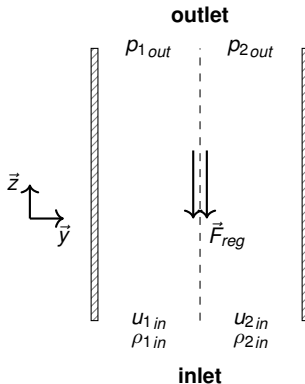
- **Mach number:** $M = \frac{u}{c} \approx 0.001 \ll 1 \Rightarrow \text{incompressible flow}$
- **Reynolds number:** $Re = \frac{\rho u D_h}{\mu} \approx 8 \times 10^6 \gg 1 \Rightarrow \text{inviscid flow}$

Appendix: Fuel assembly



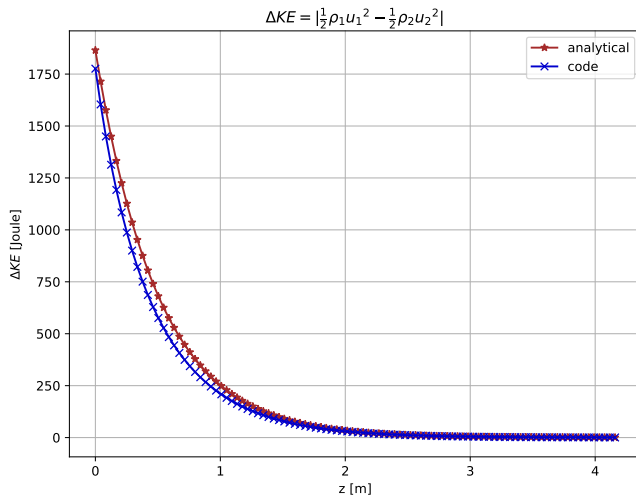


- Two parallel channels with different inlet velocities



- Analytical solution:

$$\Delta KE \equiv \left| \frac{1}{2} \rho_1 u_{1z}^2 - \frac{1}{2} \rho_2 u_{2z}^2 \right| = e^{-\frac{C_f}{D_h} z} \left| \frac{1}{2} \rho_{1in} u_{1in}^2 - \frac{1}{2} \rho_{2in} u_{2in}^2 \right|$$



Kinetic energy profile for two parallel channels



$$\text{Threshold} \equiv \frac{\sigma_{\Delta KE}}{\Delta KE} = \frac{1}{\Delta KE} \sqrt{(u_1^4 \sigma_{\rho_1})^2 + (2\rho_1 u_1^2 \sigma_{u_1})^2 + (u_2^4 \sigma_{\rho_2})^2 + (2\rho_2 u_2^2 \sigma_{u_2})^2}$$

Threshold	15.6 %
Maximum relative error	16.42 % 
Average relative error	9.27 % 

UBFC

UNIVERSITÉ
BOURGOGNE FRANCHE-COMTÉ

The CEA logo, featuring the lowercase letters 'cea' in white on a red square background. The letters are stylized with a horizontal line underneath the 'a'.

cea

Advancing the Verification and Error Analysis of Thermohydraulic Simulations in Nuclear Reactors

Master 2 CompuPhys - Internship Presentation | Ali FADLALLAH

Supervisors: Alexiane PLESSIER and Michael NDJINGA from DES/DM2S/STMF at CEA Saclay

28 August 2024